

Hamiltonian systems and symplectic methods

We have seen that the implicit midpoint rule applied to the pendulum problem is "almost conservative": the energy forever oscillates around the correct value.

In fact, this is an example of a symplectic method; any symplectic method applied to any Hamiltonian system will behave this way.

Indeed, symplectic methods satisfy an even stronger property...

Variational equation

$$u'(t) = f(u)$$

$$u: \mathbb{R} \rightarrow \mathbb{R}^m$$

$$u(0) = u_0$$

How does the initial condition u_0 influence $u(t)$? To find the answer we can solve an ODE system for the Jacobian:

$$\underline{\Psi}(t) = \frac{\partial u(t)}{\partial u_0}$$

Differentiate w.r.t. t :

$$\begin{aligned} \underline{\Psi}'(t) &= \frac{d}{dt} \frac{\partial u(t)}{\partial u_0} = \frac{\partial}{\partial u_0} u'(t) = \frac{\partial}{\partial u_0} f(u) \\ &= \frac{\partial f}{\partial u} \frac{\partial u}{\partial u_0} \end{aligned}$$

(VE)

$$\underline{\Psi}'(t) = f'(u) \underline{\Psi}(t)$$

$$\underline{\Psi}(0) = I$$

Numerical Variational Equation

Explicit Euler: $u_{n+1} = u_n + hf(u_n)$

Differentiate: $\frac{\partial u_{n+1}}{\partial u_0} = \frac{\partial u_n}{\partial u_0} + hf'(u) \frac{\partial u_n}{\partial u_0}$

This is exactly what we would get if we applied Euler's method to (VE)!

Lemma: For any RK method,

The derivative (w.r.t. u_0) of u_n is equal to the solution $\bar{\Psi}_n$ obtained by applying the same method to (VE).

Finite-dimensional Hamiltonian Systems

Coordinates: q_j

Momenta: p_j

$$1 \leq j \leq d$$

Hamiltonian: $\mathbb{R}^{2d} \rightarrow \mathbb{R}$

$$H(p, q)$$

$$p_j'(t) = -\frac{\partial H}{\partial q_j}$$

$$q_j'(t) = \frac{\partial H}{\partial p_j}$$

$$j=1, 2, \dots, d$$

$$\frac{dH}{dt} = \sum_j \left(\frac{\partial H}{\partial p_j} p_j' + \frac{\partial H}{\partial q_j} q_j' \right) = 0.$$

Setting $u = \begin{bmatrix} p \\ q \end{bmatrix}$ we have

$$u'(t) = J^{-1} \nabla H(u) \quad \text{where}$$

$$J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \quad J^{-1} = \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix} = -J$$

The variational equation is

$$\Psi'(t) = J^{-1} \nabla^2 H(u) \Psi$$

← Hessian

Example: Pendulum

$$H = \frac{1}{2} p^2 - \cos q$$

$$\nabla H = \begin{bmatrix} \sin q \\ p \end{bmatrix}$$

$$\begin{bmatrix} p \\ q \end{bmatrix}' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \nabla H = \begin{bmatrix} -\sin q \\ p \end{bmatrix}$$

$$\nabla^2 H = \begin{bmatrix} 0 & \cos q \\ 1 & 0 \end{bmatrix}$$

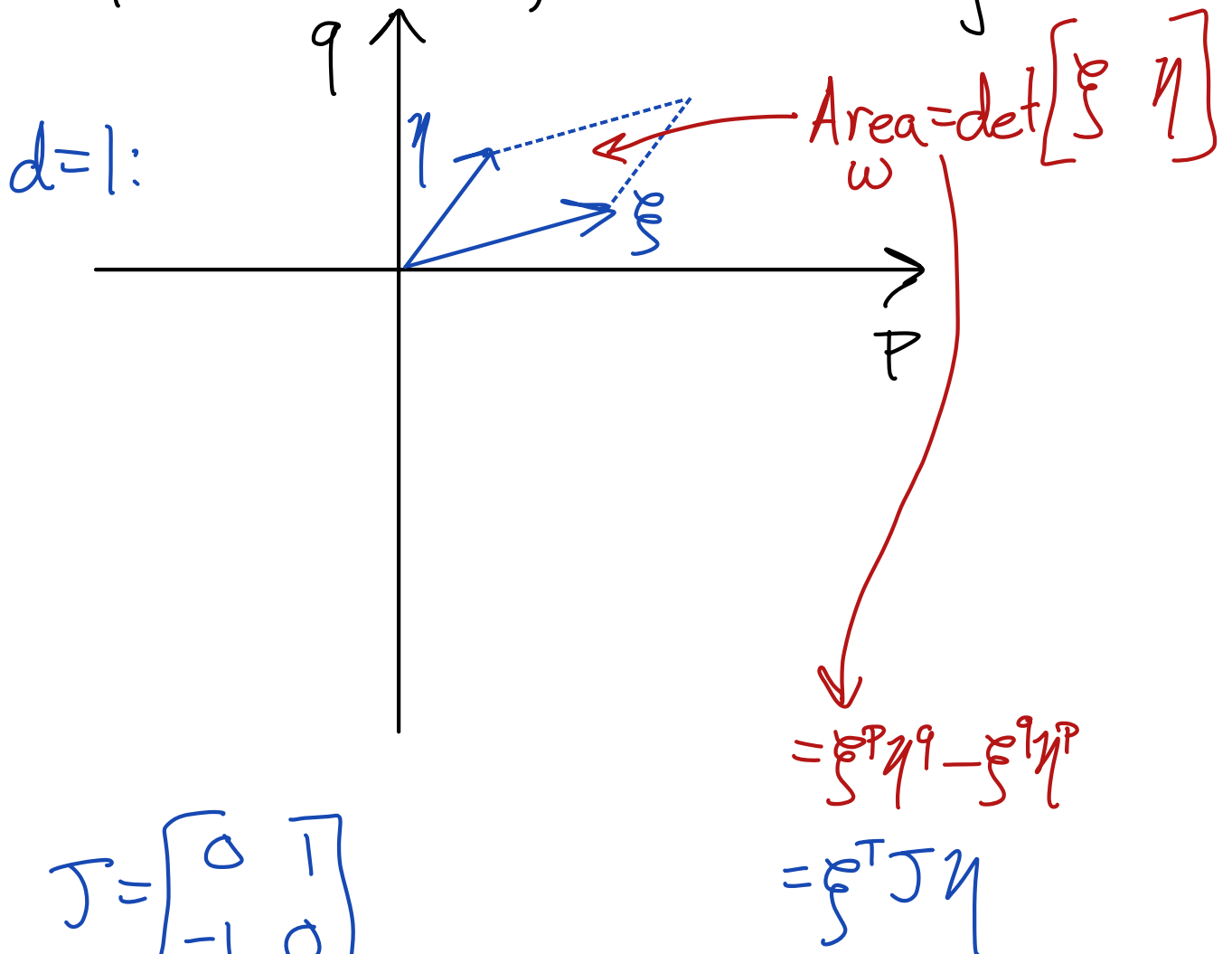
$$J^{-1} \nabla^2 H = \begin{bmatrix} -1 & 0 \\ 0 & \cos q \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} \frac{\partial p}{\partial p_0} & \frac{\partial p}{\partial q_0} \\ \frac{\partial q}{\partial p_0} & \frac{\partial q}{\partial q_0} \end{bmatrix} = \begin{bmatrix} -\frac{\partial p}{\partial p_0} & -\frac{\partial p}{\partial q_0} \\ \frac{\partial q}{\partial p_0} \cos q & \frac{\partial q}{\partial q_0} \cos q \end{bmatrix}$$

Don't
bother
with this
part

Symplecticity

A map is said to be symplectic if it preserves area, in the following sense.



Linear mapping A is symplectic if

$$(A\xi)^T J (A\eta) = \xi^T J \eta$$

$$\Rightarrow \xi^T A^T J A \eta = \xi^T J \eta \Leftrightarrow A^T J A = J$$

$$d > 1: \quad \xi = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_d \\ q_1 \\ q_2 \\ \vdots \\ q_d \end{bmatrix} \quad J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \quad I \in \mathbb{R}^{d \times d}$$

$$\omega = \xi^T J \eta = \sum_{i=1}^d (p_i^T \eta_i^q - p_i^q \eta_i^p)$$

A is symplectic if

$$A^T J A = J$$

We say a nonlinear mapping is symplectic if its Jacobian is everywhere symplectic.

Theorem 2.4 (Poincaré 1899). Let $H(p, q)$ be a twice continuously differentiable function on $U \subset \mathbb{R}^{2d}$. Then, for each fixed t , the flow φ_t is a symplectic transformation wherever it is defined.

$$\Phi_t(p_0, q_0) = (p(t), q(t))$$

We need to show that $\nabla \Phi_t$ is symplectic; i.e. $(\nabla \Phi_t)^T J (\nabla \Phi_t) = J \quad \forall t$.

Observe that $\nabla \Phi|_{t=0} = I$, so this holds at $t=0$.

Furthermore, $\nabla \phi_t = \frac{\partial u}{\partial u_0} = \mathbb{I}$ so $\nabla \phi_t$ satisfies the variational equation:

$$\frac{d}{dt}(\nabla \phi_t) = \mathbb{J}^{-1} \nabla^2 H \cdot \nabla \phi_t$$

Thus

$$\begin{aligned} \frac{d}{dt}(\nabla \phi_t)^T \mathbb{J}(\nabla \phi_t) &= (\mathbb{J}^{-1} \nabla^2 H \cdot \nabla \phi_t)^T \mathbb{J}(\nabla \phi_t) \\ &\quad + (\nabla \phi_t)^T \mathbb{J} \mathbb{J}^{-1} \nabla^2 H \cdot \nabla \phi_t \\ &= 0 \quad \text{since } \mathbb{J}^{-T} \mathbb{J} = -\mathbb{I}. \end{aligned}$$

This proof also shows that $\Phi^T \mathbb{J} \Phi$ is a quadratic invariant of the Hamiltonian system!

Symplectic numerical methods

Do numerical methods preserve symplecticity? i.e. for a Hamiltonian system do we have area preservation?

We say a numerical method is symplectic if the map $\Phi_h: u_n \rightarrow u_{n+1}$ is symplectic.

For RK methods, preservation of quadratic invariants is equivalent to symplecticity?

Theorem 4.3. *If the coefficients of a Runge-Kutta method satisfy*

$$b_i a_{ij} + b_j a_{ji} = b_i b_j \quad \text{for all } i, j = 1, \dots, s, \quad (4.2)$$

then it is symplectic.

□

Proof: This follows from two facts:

- $\nabla \phi_t^T J \nabla \phi_t$ is a quadratic invariant of the variational equation
- The numerical value of $\nabla \phi_t$ for an RK method applied to the Hamiltonian system is the same as what one would obtain by numerically integrating the variational equation.